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Batch : B3(2025-2029)

**ASSIGNMENT No: 13****TITLE:** Write a Java program to perform reading from and writing to a file using file handling operations.

## Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

## Learning Objectives

To perform file reading and writing operations using Java file handling classes.

## Theory

Java provides file handling through the java.io package. Classes such as File, FileReader, FileWriter, BufferedReader, and BufferedWriter are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.

## Code

```
import java.io.*;

public class FileDemo {

    public static void main(String[] args) {

        try {

            FileWriter w = new FileWriter("sample.txt");
            w.write("Name : Vatsal.\n");
            w.write("Welcome to SIT , Pune\n");
            w.write("Welcome to Pune\n");
            w.close();

            System.out.println("Data written successfully.\n");

            FileReader r = new FileReader("sample.txt");
```

```

BufferedReader br = new BufferedReader(r);

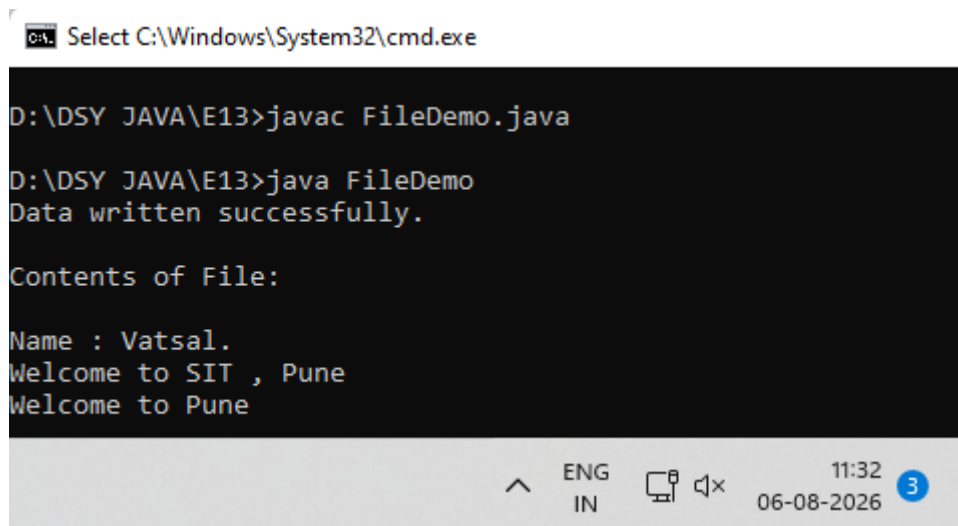
String line;
System.out.println("Contents of File:\n");

while ((line = br.readLine()) != null) {
    System.out.println(line);
}

} catch (IOException e) {
    System.err.println("File Error:"+e.getMessage());
}
}
}

```

## Output



```

Select C:\Windows\System32\cmd.exe

D:\DSY JAVA\E13>javac FileDemo.java

D:\DSY JAVA\E13>java FileDemo
Data written successfully.

Contents of File:

Name : Vatsal.
Welcome to SIT , Pune
Welcome to Pune

```

## Exercise

1. Create a program to write and read **employee details from a file**.

## Code

```

import java.io.*;

class Employee {
    int id;
    String name;

```

```

double salary;

public Employee(int empId, String empName, double empSalary) {
    id = empId;
    name = empName;
    salary = empSalary;
}
}

public class EmployeeFile {

    public static void main(String[] args) {

        Employee emp1 = new Employee(1, "Ravi", 45000.50);
        Employee emp2 = new Employee(2, "Vatsal", 55000.00);

        try {
            FileWriter writer = new FileWriter("employees.txt");

            writer.write(emp1.id + " - " + emp1.name + " - " + emp1.salary + "\n");
            writer.write(emp2.id + " - " + emp2.name + " - " + emp2.salary + "\n");

            writer.close();
            System.out.println("Data written to employees.txt successfully.\n");

        } catch (IOException e) {
            System.out.println("Error writing to file: " + e.getMessage());
        }

        try {
            FileReader reader = new FileReader("employees.txt");
            BufferedReader br = new BufferedReader(reader);

            String line;
            System.out.println("Contents of employees.txt:");

            // Reading and printing data line by line
            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();
            reader.close();

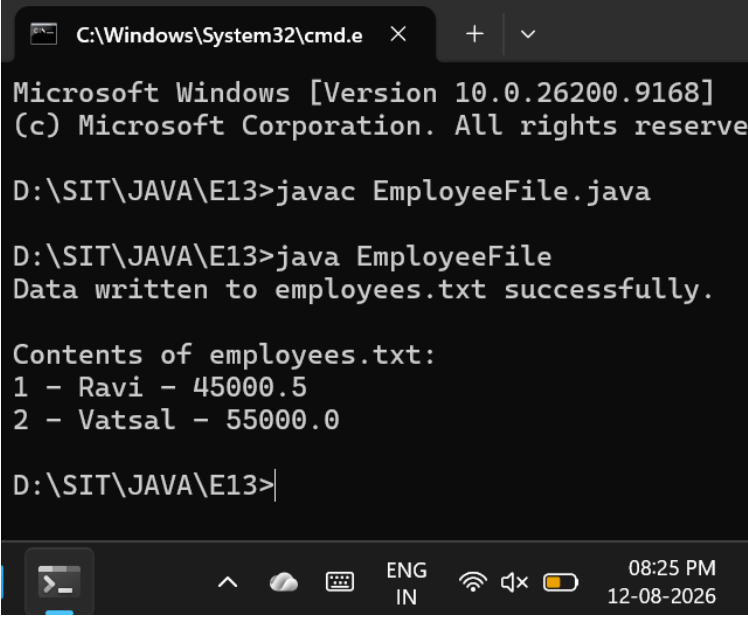
```

```

    } catch (IOException e) {
        System.out.println("Error reading from file: " + e.getMessage());
    }
}
}

```

## Output



```

C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved

D:\SIT\JAVA\E13>javac EmployeeFile.java

D:\SIT\JAVA\E13>java EmployeeFile
Data written to employees.txt successfully.

Contents of employees.txt:
1 - Ravi - 45000.5
2 - Vatsal - 55000.0

D:\SIT\JAVA\E13>

```

2. Create a Java program to write **patient details** (patient ID, name, age, and diagnosis) to a file and read the details from the file for display.

## Code

```
import java.io.*;
```

```
class Patient {
```

```
    int id;
```

```
    String name;
```

```
    int age;
```

```
    String diagnosis;
```

```
    public Patient(int pId, String pName, int pAge, String pDiagnosis) {
```

```
        id = pId;
```

```
        name = pName;
```

```
        age = pAge;
```

```
        diagnosis = pDiagnosis;
    }
}
```

```
public class PatientFile {
```

```
    public static void main(String[] args) {
```

```
        // Creating patient objects
```

```
        Patient p1 = new Patient(101, "Amit Sharma", 45, "Viral Fever");
```

```
        Patient p2 = new Patient(102, "Neha Singh", 32, "Migraine");
```

```
        try {
```

```
            FileWriter writer = new FileWriter("patients.txt");
```

```
            writer.write("ID: " + p1.id + ", Name: " + p1.name + ", Age: " + p1.age + ", Diagnosis: "
+ p1.diagnosis + "\n");
```

```
            writer.write("ID: " + p2.id + ", Name: " + p2.name + ", Age: " + p2.age + ", Diagnosis: "
+ p2.diagnosis + "\n");
```

```
            writer.close();
```

```
            System.out.println("Patient details written to patients.txt successfully.\n");
```

```
        } catch (IOException e) {
```

```
            System.out.println("Error writing to file: " + e.getMessage());
```

```
        }
```

```
        try {
```

```
            FileReader reader = new FileReader("patients.txt");
```

```
            BufferedReader br = new BufferedReader(reader);
```

```
            String line;
```

```
            System.out.println("--- Contents of patients.txt ---");
```

```
            while ((line = br.readLine()) != null) {
```

```
                System.out.println(line);
```

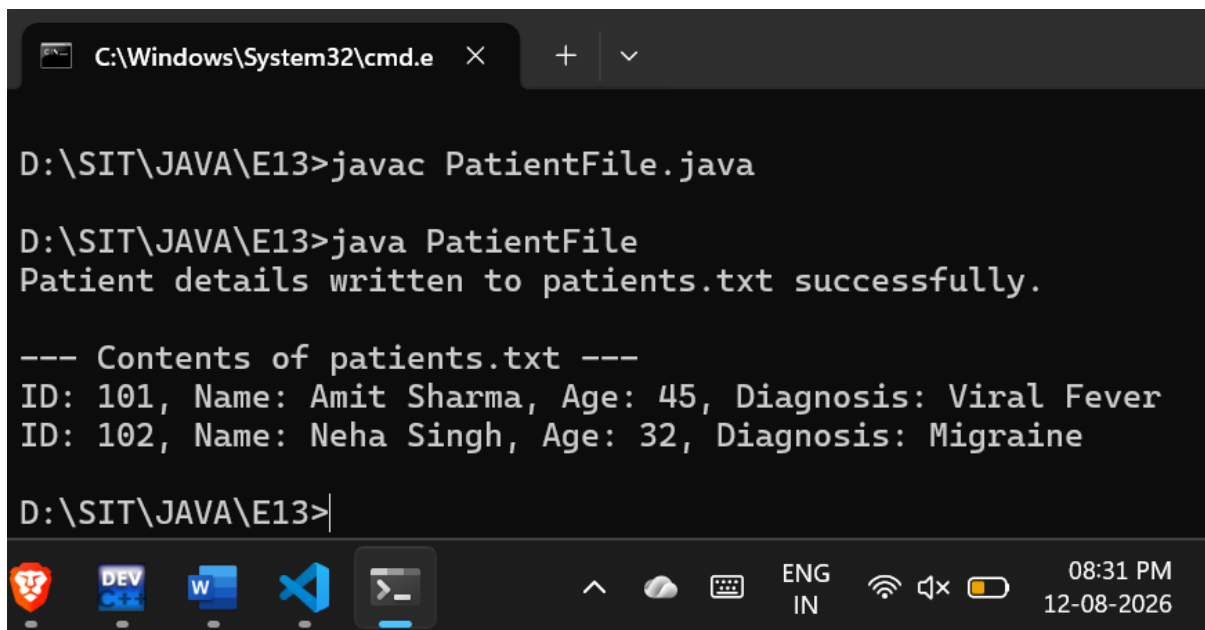
```
            }
```

```
            br.close();
```

```
            reader.close();
```

```
} catch (IOException e) {  
    System.out.println("Error reading from file: " + e.getMessage());  
}  
}  
}
```

## Output



The screenshot shows a Windows Command Prompt window with the following text:

```
C:\Windows\System32\cmd.e  X  +  v  
  
D:\SIT\JAVA\E13>javac PatientFile.java  
  
D:\SIT\JAVA\E13>java PatientFile  
Patient details written to patients.txt successfully.  
  
--- Contents of patients.txt ---  
ID: 101, Name: Amit Sharma, Age: 45, Diagnosis: Viral Fever  
ID: 102, Name: Neha Singh, Age: 32, Diagnosis: Migraine  
  
D:\SIT\JAVA\E13>|
```

The taskbar at the bottom shows icons for a shield, DEV, Word, and VS Code. System tray icons include a cloud, keyboard, language (ENG/IN), Wi-Fi, and battery. The system clock shows 08:31 PM on 12-08-2026.